

# Jacques Attinger

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## EDUCATION

### University of Chicago

*B.S. in Mathematics and Computer Science, GPA: 3.82/4.0*

Chicago, IL

*Expected May 2028*

- **Relevant Coursework:** Data Structures and Algorithms, Data Analysis With Python, Real Analysis, Linear Algebra, Probability and Statistics

## WORK EXPERIENCE

### Hemut (YC X25)

*Software Engineering Intern*

Fresno, CA

*Jun 2026 – Present*

- Built a **RAG** pipeline (Qdrant, OpenAI, Celery) that auto-bids **1000+ lane** RFP spreadsheets in parallel against historical shipment data, cutting processing time by **over 90%**
- Automated accounts receivable for a trucking TMS (Python, FastAPI, Node.js, PostgreSQL/SQL, React/TypeScript) via LLM-vision OCR document autofill and factoring exports to Excel/PDF, cutting workflow time by **over 80%**
- Deployed multi-tenant services to **AWS ECS** with Docker, automated testing, and GitHub Actions CI/CD as a forward-deployed engineer, turning weekly customer feedback into shipped features for fleets of **100+ trucks**

### Argonne National Laboratory

*Software Engineering Intern*

Chicago, IL

*Jun 2025 – Aug 2025*

- Integrated **SLADS-Net** (deep-learning dynamic sampling) from research literature with SEM control software via the PyPhenom API, increasing image reconstruction accuracy by **21%** over random sampling methods
- Built an autonomous SEM image analysis pipeline (**OpenCV**, NumPy) that acquires X-ray spectra for **500+ particles** per run without human input; adopted by multiple Argonne research groups
- Deployed a **Convolutional Neural Network (CNN)** (PyTorch) to classify energy-dispersive X-ray spectra from a Scanning Electron Microscope (SEM) into elemental classes, reducing data acquisition times by **up to 90%**

### University of Chicago Department of Physics

*Undergraduate Researcher*

Chicago, IL

*Feb 2025 – May 2026*

- Designed a safety interlock system linking lab sensors with control software to automatically shut down quantum manufacturing systems on overheating or underpressurization, saving thousands of dollars in equipment
- Developed a Tkinter **Graphical User Interface (GUI)** application to dynamically store and visualize real-time vacuum chamber pressure/temperature data
- Engineered an interactive PyQt GUI and communication layer to control a machine-vision camera and pyrometer, enabling synchronized thermal and visual data acquisition for Reinforcement Learning model training

### Princeton University Department of Mechanical & Aerospace Engineering

*Computational Research Intern*

Princeton, NJ

*Jun 2023 – Aug 2023*

- Ran Cantera simulations of e-fuel oxidation to find gaps between experiments and state-of-the-art models, leading to improved Dimethoxymethane reaction pathways
- **Publication:** Co-author, [Dimethoxymethane low- and intermediate-temperature oxidation up to 100 atm](#), *Proc. Combust. Inst.* (2024)

## PROJECTS

### ChessBuddy

*React, TypeScript, JavaScript, HTML/CSS, Vite, Anthropic Claude, LangChain, LangSmith*

- Built a React/TypeScript/JavaScript/Vite web app with a hand-built HTML/CSS interface that translates uploaded chess games into plain-English, move-by-move coaching feedback
- Designed an LLM pipeline (Anthropic Claude, LangChain, LangSmith) with JSON Schema-validated structured outputs that converts Stockfish engine evaluations into natural-language move explanations
- Implemented an LLM-as-judge evaluation pipeline, validated against human-labeled data, to guarantee explanation accuracy at scale

## TECHNICAL SKILLS

**Languages:** Python, TypeScript, JavaScript, SQL, HTML/CSS

**Tools:** PyTorch, LangChain, LangGraph, FastAPI, React, Node.js, PostgreSQL, Qdrant, Docker, AWS (ECS, S3, SQS), Celery, GitHub Actions, CI/CD, Git, GitHub, Linux, NumPy, OpenCV, Claude Code, Codex, Vercel